Runge–Kutta Methods

Geometry, slope averages, and the path to RK4

Dr. Harrington

MAD 4401 — University of Florida

Methods we already know

For $y' = f(t, y)$, approximate $y(t_n)$ by y_n.

$$\text{Euler:} \quad y_{n+1} = y_n + hf(t_n, y_n),$$

$$\text{Improved Euler:} \quad y_{n+1} = y_n + \frac{h}{2}\big[f(t_n, y_n) + f(t_n + h, y_n + hf(t_n, y_n))\big].$$

Taylor methods add higher derivatives of the solution:

$$y(t_n + h) = y(t_n) + hy'(t_n) + \frac{h^2}{2}y''(t_n) + \cdots.$$

Runge–Kutta idea: obtain higher accuracy by evaluating f at carefully chosen trial points.

A geometric starting example

$$y' = 1 + 2t, \quad y(0) = 1, \quad h = 1.$$

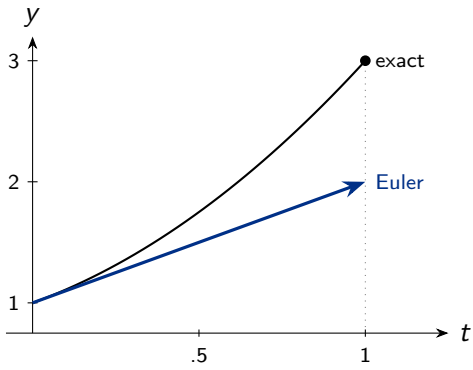
The exact solution is $y(t) = 1 + t + t^2$.

$$k_1 = f(0, 1) = 1.$$

Euler follows the initial slope:

$$y_1 = 1 + 1(1) = 2.$$

The exact endpoint is $y(1) = 3$.



The weights come from an average slope

Here the slope is $g(t) = 1 + 2t$.

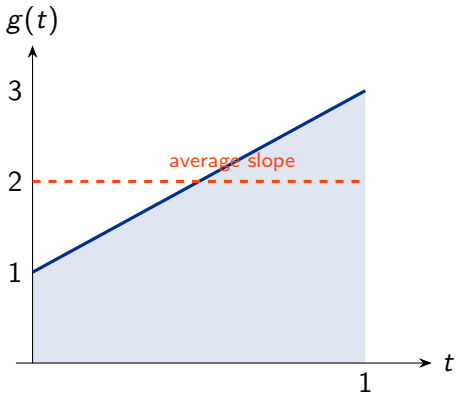
$$y(1) - y(0) = \int_0^1 g(t)\,dt.$$

The trapezoid has area

$$1 \cdot \frac{1+3}{2} = 2.$$

Thus the average slope is

$$\boxed{\overline{k} = \tfrac{1}{2}k_1 + \tfrac{1}{2}k_2 = 2}.$$



The update is $y_1 = 1 + 1(2) = 3$.

Equal endpoint weights integrate a linear slope exactly.

Improved Euler for a general ODE

Along the exact solution,

$$y(t_n + h) - y(t_n) = \int_{t_n}^{t_n+h} f(t, y(t))\, dt.$$

A trapezoid approximation would use both endpoint slopes, but the final solution value is unknown. Use an Euler predictor:

$$k_1 = f(t_n, y_n), \qquad \widetilde{y}_{n+1} = y_n + hk_1,$$
$$k_2 = f(t_n + h, \widetilde{y}_{n+1}), \qquad y_{n+1} = y_n + h\left(\tfrac{1}{2}k_1 + \tfrac{1}{2}k_2\right).$$

Improved Euler is an RK2 method. The second slope comes from a trial endpoint. For general $f(t, y)$, it is only an estimate of the exact endpoint slope.

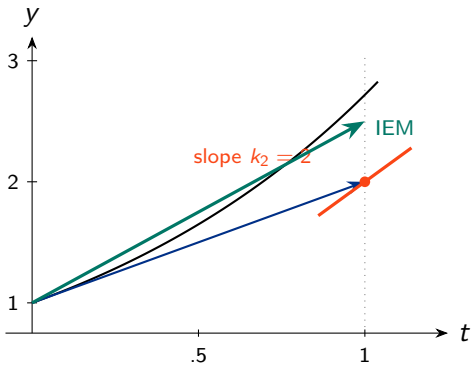
When the slope depends on the height

$$y' = y, \quad y(0) = 1, \quad h = 1.$$

$$k_1 = 1, \quad \widetilde{y}_1 = 2, \quad k_2 = f(1, 2) = 2.$$

$$y_1 = 1 + \tfrac{1}{2}(1 + 2) = 2.5.$$

The second slope is evaluated at $(1, 2)$, below the exact solution e^t. We use a large step to make the geometry visible.



The geometry of the equal weights

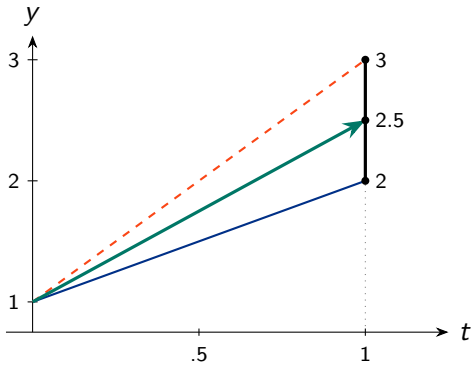
Translate both sampled slopes to the starting point $(0, 1)$.

$$1 + hk_1 = 2, \qquad 1 + hk_2 = 3.$$

The averaged slope reaches the midpoint of these two heights:

$$\tfrac{1}{2}(2) + \tfrac{1}{2}(3) = 2.5.$$

We average slopes, not angles. The translated k_2 line illustrates the average. We still evaluate k_2 at $(1, 2)$.



Midpoint RK2 samples halfway through the step

For the same example $y' = y$:

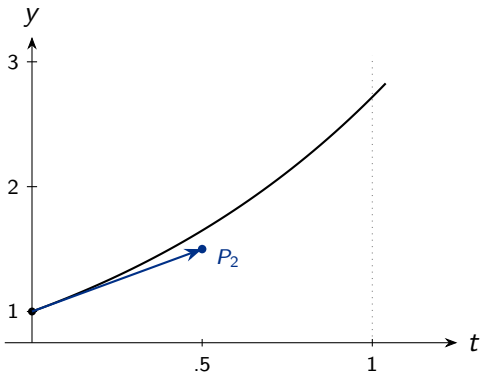
$$k_1 = 1, \quad P_2 = (\tfrac{1}{2}, 1 + \tfrac{1}{2}k_1) = (\tfrac{1}{2}, 1.5).$$

$$k_2 = f(\tfrac{1}{2}, 1.5) = 1.5.$$

Use this slope for the full step:

$$y_1 = 1 + 1(1.5) = 2.5.$$

The weights are $(b_1, b_2) = (0, 1)$. The first slope still determines the trial point.



A family of two-stage methods

Choose a fraction $c > 0$ of the step for the second sample:

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f(t_n + ch,\ y_n + chk_1), \\ y_{n+1} &= y_n + h(b_1 k_1 + b_2 k_2). \end{aligned}$$

- c chooses the location of the trial point along the Euler line.
- b_1, b_2 choose the contributions to the final slope.

These choices must work together to obtain second-order accuracy.

Geometric conditions on the weights

Write $s = (t - t_n)/h$. A constant slope must remain unchanged:

$$\boxed{b_1 + b_2 = 1}.$$

For a slope varying linearly as $g(s) = A + Bs$, its average on $[0, 1]$ is $A + B/2$. The two samples give

$$b_1 g(0) + b_2 g(c) = (b_1 + b_2)A + b_2 cB.$$

Thus the weighted sample locations must have their average at the midpoint:

$$\boxed{b_2 c = \tfrac{1}{2}}, \qquad b_2 = \frac{1}{2c}, \quad b_1 = 1 - \frac{1}{2c}.$$

Next we check why these same conditions work when f depends on y.

The connection to the Taylor method

Start a step with exact input. Evaluate f, f_t, f_y at (t_n, y_n).

$$y'' = \frac{d}{dt} f(t, y(t)) = f_t + f_y f.$$

The exact expansion is

$$y(t_n + h) = y_n + hf + \frac{h^2}{2}(f_t + f_y f) + O(h^3).$$

The second stage has expansion

$$k_2 = f(t_n + ch, y_n + chf) = f + ch(f_t + f_y f) + O(h^2).$$

Therefore the RK update is

$$y_{n+1} = y_n + h(b_1 + b_2)f + h^2 b_2 c(f_t + f_y f) + O(h^3).$$

Matching coefficients gives $b_1 + b_2 = 1$ and $b_2 c = 1/2$.

Different samples, different weights, same order

Method	c	b_1	b_2
Improved Euler	1	$1/2$	$1/2$
Midpoint	$1/2$	0	1
Ralston RK2	$2/3$	$1/4$	$3/4$

Each method has a one-step error $O(h^3)$ and global error $O(h^2)$ under the usual smoothness and Lipschitz assumptions. Their errors need not be equal. The agreement for $y' = y$ is a special feature of that example.

What RK2 leaves out

For $y' = y$, any member of this RK2 family gives

$$y_{n+1} = y_n \left(1 + h + \frac{h^2}{2}\right).$$

But the exact step is

$$y(t_n + h) = y(t_n) \left(1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24} + \cdots\right).$$

Taylor methods compute higher derivatives explicitly. **RK4 uses four coordinated slope evaluations to match through fourth order.** Additional slopes alone do not guarantee a higher order.

RK4: the first midpoint trial

Continue with $y' = y$, $y(0) = 1$, $h = 1$.

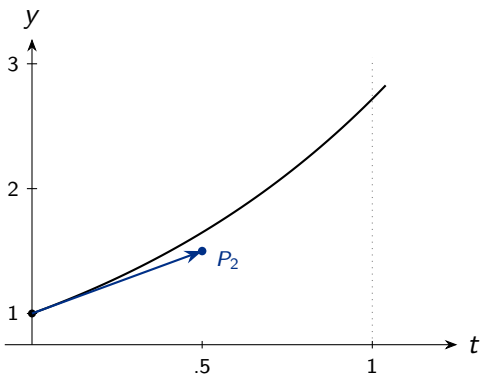
$$k_1 = f(0, 1) = 1.$$

Move halfway along the first slope:

$$P_2 = (\tfrac{1}{2}, 1 + \tfrac{1}{2}k_1) = (\tfrac{1}{2}, 1.5).$$

$$\boxed{k_2 = f(P_2) = 1.5}.$$

The black curve is e^t. The blue arrow constructs the trial point.



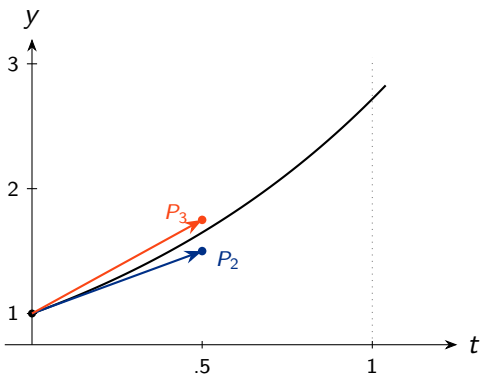
RK4: a revised midpoint trial

Use k_2 to form another trial point, measured from the original start:

$$P_3 = (\tfrac{1}{2}, 1 + \tfrac{1}{2}k_2) = (\tfrac{1}{2}, 1.75).$$

$$\boxed{k_3 = f(P_3) = 1.75}.$$

Both midpoint trials have the same time coordinate but different heights. We evaluate f again because the height can affect the slope.



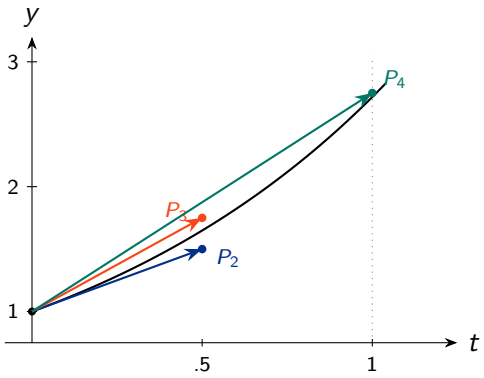
RK4: an endpoint trial

Use the revised midpoint slope for a full trial step from $(0, 1)$:

$$P_4 = (1, 1 + k_3) = (1, 2.75).$$

$$\boxed{k_4 = f(P_4) = 2.75}.$$

The trial endpoint supplies one more slope.
The final answer will combine all four slopes.



RK4: the final weighted slope

$$\overline{k} = \frac{1}{6}k_1 + \frac{1}{3}k_2 + \frac{1}{3}k_3 + \frac{1}{6}k_4.$$

For our example,

$$\overline{k} = \frac{1 + 2(1.5) + 2(1.75) + 2.75}{6} = \frac{41}{24}.$$

$$\boxed{y_1 = 1 + \overline{k} = \frac{65}{24} \approx 2.708333}.$$

The trial endpoint was 2.75. The exact endpoint is $e \approx 2.718282$. The weights sum to 1.

Each midpoint slope receives twice the weight of either endpoint slope.

The classical RK4 algorithm

$$
\begin{aligned}
k_1 &= f(t_n, y_n), \\
k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \\
k_3 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right), \\
k_4 &= f(t_n + h, y_n + hk_3), \\
\boxed{y_{n+1}} &\boxed{= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)}.
\end{aligned}
$$

Every trial height is measured from y_n. Here the k_i are slopes, so the update multiplies their weighted average by h.

A geometric explanation for the RK4 weights

First consider $y' = g(t)$, so trial heights do not affect the slopes. The sample times are $0, 1/2, 1/2, 1$ in normalized coordinates. Choose symmetric weights (a, b, b, a). Matching the average of 1 and s^2 on $[0, 1]$ gives

$$2a + 2b = 1, \qquad \frac{b}{2} + a = \frac{1}{3}.$$

Solving yields $a = 1/6$ and $b = 1/3$. These weights also integrate s and s^3 exactly.

$$\frac{1}{6}g(t_n) + \frac{2}{3}g(t_n + h/2) + \frac{1}{6}g(t_n + h)$$

This is Simpson's average-slope formula. For general $f(t, y)$, the trial-point formulas must also satisfy Taylor order conditions. The quadrature analogy alone is not a proof of order four.

A Taylor check on our geometric example

For $y' = y$ and an arbitrary step size,

$$\begin{aligned} k_1 &= y_n, \\ k_2 &= y_n(1 + h/2), \\ k_3 &= y_n(1 + h/2 + h^2/4), \\ k_4 &= y_n(1 + h + h^2/2 + h^3/4). \end{aligned}$$

Substitution gives

$$y_{n+1} = y_n \left(1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24}\right).$$

This matches the fourth-degree Taylor expansion of $y_n e^h$. This calculation verifies the matching for $y' = y$ specifically.

Comparison for one step of size one

For $y' = y$, $y(0) = 1$, the exact value is $y(1) = e$.

Method	Approximation	Absolute error
Euler	2.000000	0.718282
Improved Euler	2.500000	0.218282
Midpoint RK2	2.500000	0.218282
Classical RK4	2.708333	0.009948

For this equation, Taylor methods of degrees two and four give the same updates as RK2 and RK4, respectively. This equality is not true for arbitrary ODEs.

An example with both time and height dependence

$$y' = -2ty^2, \qquad y(0) = 1, \qquad h = 0.2.$$

Stage	Trial point (t, y)	Slope $f(t, y)$
k_1	$(0, 1)$	0
k_2	$(0.1, 1)$	-0.2
k_3	$(0.1, 0.98)$	-0.19208
k_4	$(0.2, 0.961584)$	-0.3698575156

$$y_1 = 1 + \frac{0.2}{6}(0 - 0.4 - 0.38416 - 0.3698575156) \approx \boxed{0.9615327495}.$$

The exact solution $y(t) = 1/(1 + t^2)$ gives $y(0.2) \approx 0.9615384615$.

Local and global error conventions

Let y^*_{n+1} be one numerical step starting with the exact value $y(t_n)$.

$$d_{n+1} = y(t_n + h) - y^*_{n+1}.$$

Method	One-step error d_{n+1}	d_{n+1}/h	Global error
Euler	$O(h^2)$	$O(h)$	$O(h)$
RK2	$O(h^3)$	$O(h^2)$	$O(h^2)$
RK4	$O(h^5)$	$O(h^4)$	$O(h^4)$

Some texts call d_{n+1}/h the local truncation error. Always check the definition. Global orders assume sufficient smoothness and a Lipschitz condition in y over a fixed time interval as $h \to 0$.

What to carry forward

- Euler uses the slope at the start of the step.
- Improved Euler is RK2 with equal weights on the initial and predicted endpoint slopes.
- Midpoint RK2 uses the first slope to locate a trial midpoint, then uses the midpoint slope for the update.
- RK4 uses two midpoint trials and an endpoint trial, then combines four slopes with weights $1/6, 1/3, 1/3, 1/6$.

Taylor expansions determine the accuracy. Geometry explains where the slopes are sampled and how they are combined. Adaptive RK2/3 and step-size selection belong to the next presentation.